

Gender Gaps in CS1 Exams: The Role of Biased Question Framing

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Abstract

CS education is shaped by gendered cultural norms, and differences in assessment performance between men and women persist in many countries. Over multiple years, we observed noteworthy and significant, although inconsistent, differences between male and female students in our CS1 course's exam grades. We build on previous research that performance differences between men and women result from repeated interaction with biased learning environments rather than individual ability differences. In our context, we query gender bias in the design of high-stakes exams. Our hypothesis is that use of real-world settings of programming questions that are perceived as gendered amplify exclusion by embedding stereotype-evoking contextual cues.

We conducted two experiments to evaluate the effect of gendered question framings on student performance during trial exams. In the first experiment, three parallel versions of the exam were created, containing exclusively masculine, feminine, or gender-neutrally framed questions. Results showed that the fully masculine exam design increases the performance gap between men and women. In the second experiment, masculine and feminine framed question topics were blended evenly within a single version of the exam, and the effect disappeared.

Our results suggest that while gendered framing at the level of individual assessment tasks may be too subtle and transient, their consistent presence across a whole exam can meaningfully affect performance. We point to the need for systematic approaches for assessment design which take into account how unexamined real-world task framings can privilege men and advocate for assessment methods that mitigate gender inequity in CS education.

CCS Concepts

• **Social and professional topics** → **Student assessment; Gender; Computing education.**

Keywords

CS1, assessment design, gender, gender bias, real-world assessment

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1 Introduction

Today's technological progress is often promised to benefit society as a whole without exception. However, ensuring that benefits of technology are shared is a key challenge for historically male-dominated fields, such as Science, Technology, Engineering, and Mathematics (STEM). Women are underrepresented in science and engineering [14], which results in prioritization of men's needs and silencing of women's voices [13]. This emphasizes the importance of inclusive education for STEM disciplines and Computer Science (CS) as one of the pivotal goals of global sustainable development [39]. The issue is especially acute for CS, where men continue to make up the overwhelming majority of graduates in information and communication technologies (ICT) bachelor programs in OECD countries [31].

Gender differences in exam performance are not easily explained by ability, as girls often perform better or similarly to boys in mathematics and sciences at primary and secondary school [5, 26, 30]. It is sociocultural factors, such as learning environments, attitudes to digital skills and confidence in one's skills, that primarily affect educational choices and career aspirations [15]. Today, no country can claim to have achieved full gender equality. In Norway, as in other countries understood to have made significant gains regarding gender equality [37], there are still notable and concerning gender inequalities in enrollment to CS study programmes and employment in academic positions [27, 35, 36]. Despite universities' attempts to address these challenges with strategic plans on inclusive and accessible learning environments [40, 41], these types of equity documents are often aspirational rather than reflecting actual institutional practices [1].

Fig. 1 shows the existing gender gap¹ in results of the last few years across multiple introductory CS courses (CS1) in Norway. The courses we look at are open to students from across the natural sciences. The initial motivation for this study was the observation of a varying gender gap in the INF100 course at the University of Bergen. Between 2020 and 2021, the gap had increased from 0.36 to 1.06 in mean grades, even though both courses were held by the same lecturer, taught remotely (due to the pandemic), had the same exercises, and used the same assessment form of a single final exam.

The assessment design of these high-stake exams may be one of the biggest contributing factors to the observed gender differences in performance. Although it is impossible to identify a single factor, we believe that the design and the format of the exam could be key contributors. High-stakes examinations in large, obligatory courses

¹Throughout our study, students' gender information is based on the binary legal gender M/F that they have recorded in the Norwegian Population Register. It is common in Norwegian academic practice to use this classifier for collecting statistics about staff and students. Since 2016 legal gender can be self-identified in Norway within the existing two-gender registration system [29]. We acknowledge that this does not reflect students' full spectrum of gender identity.

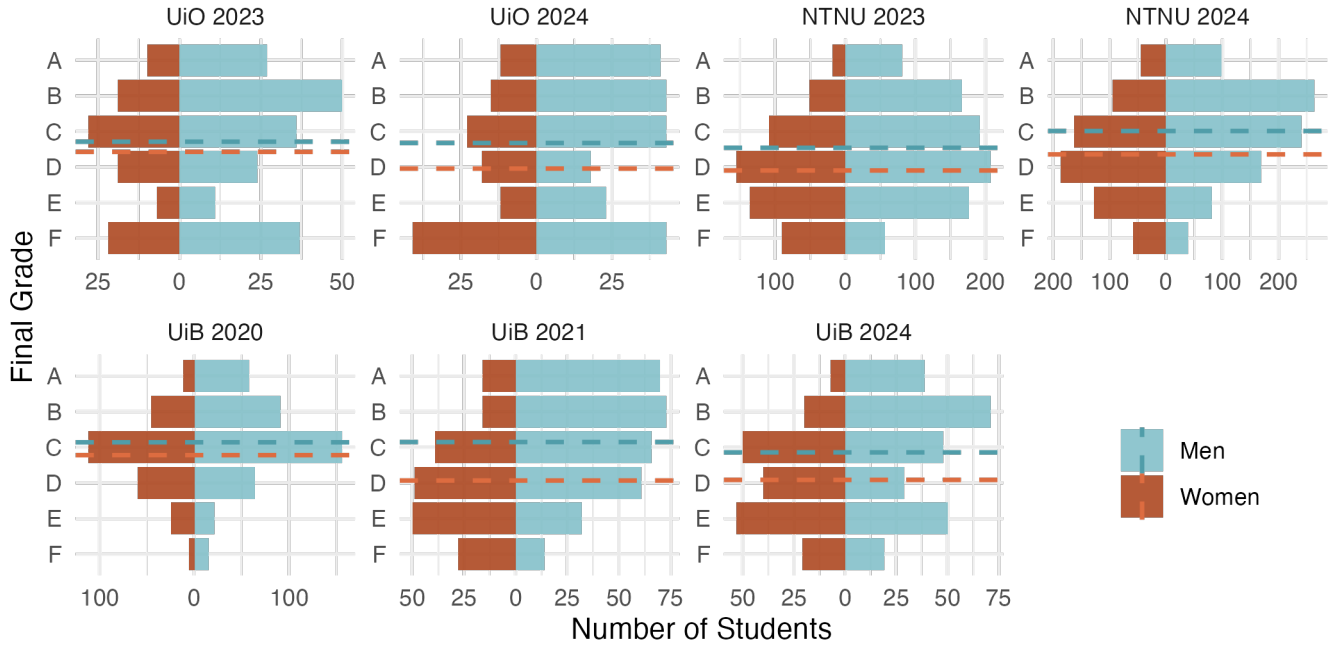


Figure 1: Final grades in selected CS1-like autumn courses for a natural sciences audience across Norway (UiO IN1900, NTNU TDT4110, UiB INF100). We see men doing better across the board, with occasional large variations in the size of the gap from year to year, even where we know that course conditions were comparable, such as UiB in 2020/2021, where the gap in mean grade widened from 0.36 letter grades to 1.08 grades. [28].

make an already performance-focused learning environment more sensitive to unexamined biases. Questions are often embedded in real-world topics, which is used to provide more authentic modes of assessment [2]. Instructors drawing on their interests and experiences (and what they might presume to be the interests of students) to write assessments may not seem problematic. However, their choice may draw on experiences and interests of instructors who come from groups that are already expected to succeed in CS. Design of effective real-world assessments is challenging: the more integral detailed knowledge of a real-world context is for interpreting the task, the greater the risk that students will misunderstand the task. Put simply, if students know little about football, they may misunderstand a CS task that uses football-specific content in its description. As a result of existing gender imbalance in the field (particularly among instructors) and limited institutional resources for exam design (instructor diversity, time, and training) this approach to assessment design risks reinforcing gender inequality. If CS assessments frequently or exclusively use examples that are more relatable to men, assessment design can privilege those students who are already overrepresented in the field.

With this study, we aim to answer the following question: *Does gendered framing of exam tasks have an effect on the gap between men and women in CS1 performance?*

We present the results of two controlled learning experiments that were conducted on the INF100 course’s trial exams in autumn 2024 and spring 2025 at the University of Bergen. In these studies, we created intentionally gendered topics and moderated how they

were included in exam questions, to see their effect on the performance gap. Each study is reported here individually with its own design, participant description, and results. The study reports are preceded by a shared description of materials and experimental procedures, and followed by a joint discussion and a section on limitations, including selection bias considerations. Together, these studies inform ongoing discussions on assessment design choices and their implications for gender equity in CS education.

2 Related Works

CS and ICT are still coded masculine in most of the world [10, 37]. This phenomenon is well studied in relation to educational choices. In Norway, young women find CS attractive due to its importance and opportunities it provides, but report little motivation coming from personal experience, sometimes naming the field “boyish” and “geekish” [11, 34]. Research from STEM fields in Scandinavia shows that gender bias may also contribute to gender differences in drop out rates, class participation and performance [3], to insecurity in abilities [17], and lowered sense of belonging [19]. Although boys and girls perform similarly in mathematics and sciences at primary and secondary school [26, 30], by the time they reach post-secondary education, women report polarized experiences within and attitudes towards CS, and experience barriers to study and work in the field [10].

Addressing gender equity in computer science has been approached from varying perspectives. Corneliussen [10] distinguishes two theoretical approaches for interrogating gender inequality in

CS: an approach informed by scholarship from gender studies concerned with how we “do gender” [32], and an approach from social psychology that interrogates how gender biases impact girls and women, focusing on concepts such as *sense of belonging* [24]. While Corneliusen’s work draws on gender studies theories, much of the scholarship in Norway about gender equity in STEM is informed by social psychology. What both theoretical framings share, is a concern with how sustained exposure to biased environments is consistently named as a contributor to the reproduction of gender inequality in CS.

What is it in the CS learning environment that conveys messages of exclusion? Many learning design choices are informed by individual instructors’ experiences, standards and preferences, and thus can contribute to gender inequality. Evidence from online learning research indicates that observable gender differences in preference and performance can be triggered both through visual and textual cues [22, 25, 33, 38]. In non-digital settings, messages of women’s incompatibility with the subject can also be communicated through physical objects in the learning environment [7].

Knowing that assessment and course design can impact gender equity, identification of how biases emerge in assessments is essential. Assessment is a fundamental part of the learning experience and has the highest risk of negative consequences: students’ interpretation of the final score serves as confirmation that people like them can or cannot succeed in the field [20]. This is especially relevant for high-demand CS1-level courses with high-stakes final exams. Evidence shows that this specific assessment form can widen the gaps between student groups, contributing to educational inequity [16].

Programming exams often combine a short-answer section with more complex programming problems. Coding tasks commonly get inspiration from authentic assessment, the practice of integrating real-world framing topics into the task to test students’ problem-solving and knowledge application skills [2, 9]. The selection of those scenarios by teachers can therefore create explicitly or implicitly biased assessments. By either misinterpreting signals of cultural capital as evidence of ability [6, 21], or as a result of implicit bias and belief in stereotypes about which group is more competent [4], teachers might bias their evaluation of assessed knowledge.

Despite its importance, identification of stereotype-evoking cues in examinations remains a challenging task. There is limited empirical research in this area due to methodological limitations, including weak reliability of estimands, underpower and low external validity [18], as well as ethical considerations of running experiments during summative assessments.

3 Overview of both studies

In the studies below, we tested how textual cues can have an impact on students’ performance during CS exams. For this purpose, we constructed gender biased CS problems and compared students’ performance on these problems. For clarity in the rest of the paper, we distinguish the *Concept* and the *Topic* of a question:

Concept: the CS content area being assessed in the problem that was moderated; chosen such that the general set of assessed concepts aligned with previous exams.

Topic: real-world setting of the problem that the concept was textually embedded into; selected to intentionally frame the concept with a topic that has a known gender bias.

Typically, instructors for CS1 identify CS concepts that are to be assessed in the exam questions, and then embed these concepts in relevant real-world topics. This practice has been commonplace in CS1 exams at UiB. Previous CS1 exams have for example used references to football leagues or political events or board games. For both our studies, a concrete list of topics was selected based on the results of a prestudy (described below). The topics that the students in the prestudy perceived as the most gendered (masculine, feminine) or most gender neutral were identified. Using these topics, exam questions were then crafted and categorized as masculine, feminine, or gender-neutral based on their topic framings. These problems were subsequently used to design the trial exams described below.

Prestudy. In order to verify that students’ perception of topics being attributed to a certain gender aligns with ours, we ran a prestudy in which we surveyed 194 students evenly distributed across genders and study programs at UiB. Our motivation was to understand how students interpreted topics with regard to gender and to identify strongly gendered topics for use in assessment design. In this survey, respondents were asked to rate 50 real-world topics based on two criteria: “How masculine/feminine is the topic?” (7-point scale: -3 = completely masculine, 0 = neutral, 3 = completely feminine), and “How familiar are you with the topic?” (5-point scale: 1 = not familiar, 5 = highly familiar). We asked about familiarity because we do not assume that men and women are necessarily personally interested in certain topics, but we sought to understand if certain topics are more readily associated with men or women. Based on the results of the survey, we chose the topics perceived as most masculine (Football Leagues & Rules, Gas Stations, Carpentry, Star Trek), most feminine (Acne Products, Poetry, Hotels / Hospitality) and most gender-neutral (Fruit, Smartphones, Sodas) for our study design.

Course Context. This study presents the case of INF100, the largest bachelor level course at the University of Bergen (UiB), compulsory for all students studying in the STEM faculty. The objective of the course is to provide a thorough and practical introduction to programming, using Python as its programming language. The course is taught in Norwegian and is taken by almost one thousand students annually. At the end of the course, students are assessed with a 4-hour final exam that counts towards 80% to 100% of the course grade. The exam typically consists of an automatically graded part (multiple choice and short code snippets) and a manually graded part (short essays, code writing). Each part represents approximately half of the exams’ total points. A few weeks before the final exam, a *trial exam* is run as a voluntary activity. Here, students get a chance to test their disciplinary knowledge and can experience real exam conditions for the first time in their bachelor degree. The design and duration of the trial exam matches the final exam, with the only difference that it is not graded for students, and that students are allowed to leave at any time.

Experiments. Inspired by [7, 8], we conducted two controlled educational design experiments during the INF100 course’s trial

exams in autumn 2024 and spring 2025. We understand that the voluntary nature of the trial exam raises concerns of selection bias, and discuss this point further in the Limitations section. However, as educators, we hold an ethical responsibility to students to investigate assessment design in a trial exam setting before implementing changes in the final exams.

Both iterations followed a moderation-of-process experimental design: we measured how the framing of topics in assessment design (independent variable) affects performance at the trial exam (dependent variable), and how the strength of the effect depends on gender (moderator). To screen out students who were minimally engaged (e.g. students who came to read the questions, but did not complete the exam), exam attempts shorter than 75 minutes were dropped. Nobody in that time bracket achieved more than half of the possible score.

The experimental design varied between the two semesters. Study One implements a between-subject approach to investigate the effect of encountering a completely gender biased trial exam, while Study Two utilizes a within-subject approach to investigate the effect of mixing (within one single trial exam) questions that are biased to different amounts.

The data and analysis scripts are available as a Zenodo repository: <https://zenodo.org/records/19829260>.

4 Study One

We created three versions of the trial exam, which were identical in technical content and assessed CS concepts, but different in the topic of the questions. Each version was specifically constructed around a set of topics that the prestudy showed to be seen as gender biased (masculine, feminine) or gender neutral.

Our initial hypothesis was that gendered exam design will affect the gap in performance, and that the effect's size would be associated with student's gender: we expected the feminine-topic version of the exam to produce higher scores for women compared to the masculine-topic version, while men's performance would not be impacted.

4.1 Method

The experiment was run during the trial exam in the Autumn 2024 semester. 75% of the automatically graded problems were identical in the three exam versions, while the remainder and all of the manually graded questions were wrapped in gendered topics. Students were randomly assigned one of the three versions of the trial exam. The exam answers were assessed by two of the paper's authors.

The trial exam was taken by 222 students. Over a third of students are enrolled in CS, the remaining students are enrolled in other STEM programs such as earth sciences, chemistry and engineering. Students were aware that they would be assigned different variations of the trial exam, but did not know other details of the experiment. Although they were free to leave at any point of the exam, engaged participation was strongly encouraged before and during the trial exam. Of the initial 222 students, 183 students passed the 75 minute time cutoff and were included in the analysis (95 women, 88 men).

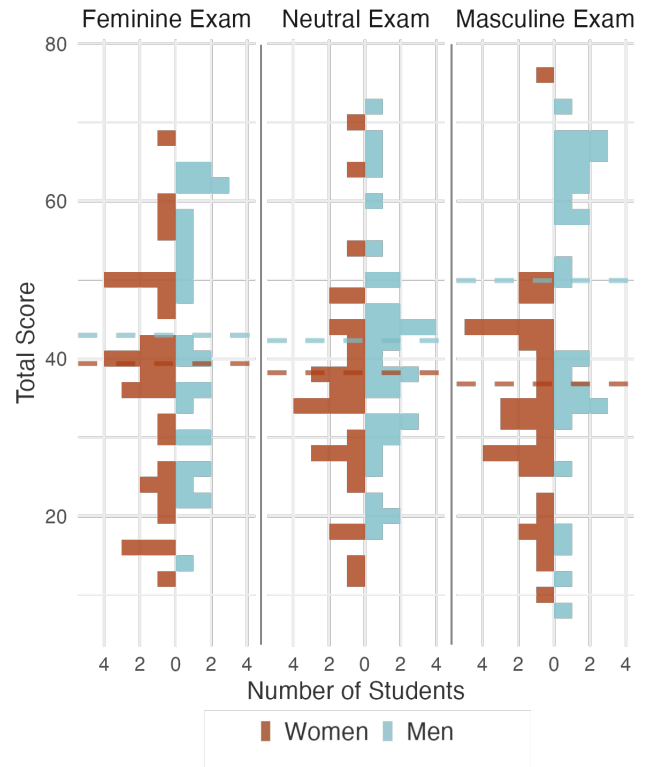


Figure 2: Distribution of points across the three versions of the trial exam for students with an attempt longer than 75 minutes. Horizontal lines indicate mean score for the corresponding gender. The masculine exam version shows a significant gap between genders.

4.2 Results

Overall, we observed a gender gap in exam performance. Men score higher than women on average across the trial exams, with the masculine version showing the largest gender gap. A $2(\text{gender}) \times 3(\text{version})$ ANOVA analysis (R function `stats::aov`) has identified a significant mean scores difference between genders, when averaged across the three versions ($F(1, 177) = 11.54, p = 0.00084$), with men scoring higher overall. Averaged across student gender, no version of the exam shows significantly better scores than another. This confirms that adapting the framing story did not make the versions more or less difficult overall.

For the analysis of specific gender-by-version contrasts, pairwise comparisons of all gender-version combinations were conducted. Post hoc tests using the Tukey HSD test (R function `stats::TukeyHSD`) showed one pair with a significant difference: in the masculine-framed versions, men score higher than women ($p_{\text{adj}} = 0.0106$). No other combination showed a significant difference.

How gender moderates the effect of assessment design on this score gap contradicts our initial hypothesis. Our assumption that women will benefit from more feminine topics was not shown. Instead, results show that the gap is increased under a masculine

assessment design while the neutral or feminine versions only show insignificant differences.

5 Study Two

The first study utilized a classic control/treatment assignment. While this method is widely used for educational experiments, it cannot, for ethical reasons, be implemented in real high-stake assessments. In the second study, we therefore attempt to derive an exam format that can be used during final exams where all students need to be presented with the same exam paper. We drew inspiration from conjoint experiments in HCI and UX user studies and built a fully-crossed experimental design that allows us to study biased framing of individual problems, rather than a completely biased exam design as in Study One. Our assumption was that spreading differently biased topics over the exam will not neutralize the perception of individual problems as being gendered. Therefore, any gaps in performance will still be differently sized when looking at the masculine and feminine subsets of the exam questions.

Our hypothesis for the second study was that gendered framing of exam tasks also has an effect on the gap between men and women in CS1 performance on an item-by-item level.

5.1 Method

The second iteration was designed as a within-cohort factorial experiment. All participants belong to one cohort, with conditions varying within the factors matrix rather than across cohorts. To reduce the risk of systematic confounding, we added three control variables. Hence, the automatically graded part utilized $2 \times 2 \times 2 \times 2$ fully crossed manipulations of the independent variable, the topic's gender bias (masculine / feminine), with question types (multiple choice / return statement), preparation asymmetry (familiar tasks / new tasks) and the place in the curriculum in relation the introduction of conditions and loops, which are significant threshold concepts (before / after). In the manually graded part, we distinguished only question type (explanation essay / coding task) and the topic's gender bias. Every question in Study Two's trial exam was thus given either a masculine or a feminine topic framing, again taken from the topic lists identified in the prestudy. The questions in each part were presented in a randomized order.

The trial exam took part in spring 2025 and was taken by 157 students. Notably, in the spring semester, students come from more heterogeneous disciplinary backgrounds: the majority come from biology, followed by medical technologies, nanotechnologies, and geoinformatics programs. This variation in student background creates an additional confounder in the study that makes spring and autumn cohorts harder to compare directly (see Limitations section). As in the previous semester, participation was strongly encouraged before and during the trial exam. This time, we provided some baked goods and fruit in the middle of the trial exam to motivate students to stay longer, and hence engage more with the trial exam. Of the initial 158 students, 146 students' exams were included in the analysis (97 women, 49 men).

5.2 Results

Study 2 results revealed no visible differences in gender performance under biased individual questions. In contrast with Study

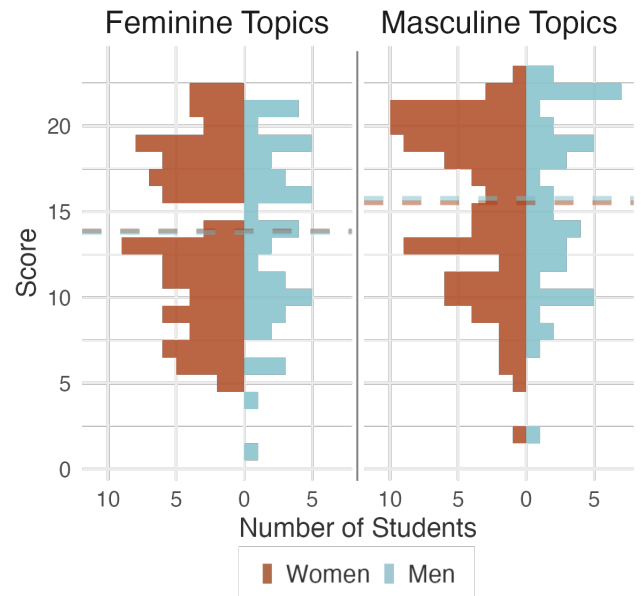


Figure 3: Distribution of points across masculine- and feminine-framed questions of the trial exam for students with an attempt longer than 75 minutes. Horizontal lines indicate mean score for corresponding gender.

1, we needed to account for between-student variance (as they each solved both masculine and feminine questions). A repeated-measures ANOVA (afex: :aov_ez) revealed no significant main effect of gender ($F(1, 143) = 2.80, p = .097$), nor of topic bias ($F(1, 143) = 0.24, p = .628$), but a significant gender-topic interaction ($F(1, 143) = 5.76, p = .018$). However, this interaction did not show in the item-level analysis. A cross-classified linear mixed model (lme4: :glmer) with students and questions as crossed random effects ($N = 3190$ observations; 145 students, 22 questions) showed no significant main effects of neither gender ($p = .748$), topic bias ($p = .808$), nor their interaction ($p = .134$). Average marginal effects of topic bias, estimated separately by gender, were overlapping and non-significant for both women ($\Delta = +2.3$ pp, 95% CI $[-15.9, 20.5]$) and men ($\Delta = +2.3$ pp, 95% CI $[-15.9, 20.5]$).

6 General Discussion

This work examined whether the embedding of CS exam questions into topics that are perceived as gender biased influences students' performance. Across two studies with complementary experimental designs, we find indications that gender-biased framings only produce detectable effects when bias is experienced consistently throughout the assessment. This indicates that the effect can be too subtle and transient on an individual item level.

Taken together, the results highlight the importance of further investigations into the diversity of assessment practices in CS1. Authentic, real-world problem contexts may not be inherently problematic, provided instructors frame the topics in ways that do not reproduce existing biases in CS. We were encouraged by the possible applicability of the second, fully crossed experimental design

to graded summative assessments. We must emphasize that addressing gender biases in assessment design requires more than simply adding counter-stereotypical examples, and demands an attention to how gendered topics are selected and used in distributed assessments.

Importantly, we encourage caution, as how topics are perceived in terms of gender is highly specific. Our use of the prestudy data was an intentional strategy for addressing our own biases about how gendered different topics are. If instructors choose topics from their (limited) frame of reference, it may result in a biased representation of real-world topics in CS education. This often may happen unintentionally, since instructors choose topics that they know well enough to build up a meaningful exam question, and many CS instructors are from groups that are already well represented in the field (women make up less than 30% of new professors in technology [23, 27]). Unexamined assessment design practices thus result in the design of assessments that privilege men (because of their probable familiarity with topics that the examiner was likely to have picked).

Designing CS assessments with real-world topics that reflect the experiences of young women demands an understanding of students' interests and experiences. This poses a real challenge for instructors that needs to be overcome. In our research, we conducted the prestudy to identify relevant possible gendered topics for use in our experiments. For instructors, finding similar processes for the identification of real-world examples that are relevant to all students may be unfamiliar and challenging.

7 Limitations

The first concern about the external validity of the study findings is the formative and voluntary nature of trial exams. We targeted trial exams to balance the experiment design affordances that the format provides, with the ethical considerations of running educational experiments during summative assessments. Self-selection may influence observed outcomes. Students' motivation or confidence (which are shaped by many aspects of identity and experience) may impact their decision to attend trial exams. Students who opt to attend trial exams may be disproportionately those who doubt or question their ability. To mitigate possible self-selection bias, we actively promoted the trial exam through regular announcements and the provision of free food halfway through the trial exam. From a post-survey, we do know that students who attended the second trial exam reported relatively high self-estimations on how seriously they took this trial exam, suggesting that students treated the trial exam as a realistic experience of a final exam.

A second concern was differing rates of participation from men and women enrolled in the course. We observed that women participated at higher rates in both of the trial exams: in Study 1, 64% of the enrolled women in the course participated in the trial exam, while only 33% of the enrolled men did. In Study 2, 67% of the enrolled women and 40% of the men participated in the trial exam. This reproduces anecdotal observations from other optional activities in our undergraduate courses, where women are more likely to participate than men. It is difficult to mitigate or redress this limitation in the research as it reflects broader gender norms where

women typically agree to volunteer or take part in optional activities more often. To redress this, it is thus necessary to investigate performance on mandatory assessments where participation will not rest on individual motivation or social expectation. Knowing this, we intend to compare our findings using the balanced assessment design from Study 2 with another iteration using a future *final* exam of INF100.

Another limitation of the study concerns the degree to which the gendered framings were even noticeable enough to be observed by the students during the trial exams. The prestudy provided us with estimates of a topic's gendering by students at the university. However, it is challenging to ensure that gendered framings were strong or clear enough to consistently be interpreted as masculine/feminine by students. A framing topic that is a natural embedding for the programming concept (sort the teams in a sports league table) may be perceived more strongly than an incidental topic embedding (sort a list of holiday destinations/fruit/teams by name). In introductory CS problems, the complexity and level of abstraction of some concepts also makes it difficult to embed them in one of the real-world topics from the prestudy while preserving the clarity of the task at hand. This creates a tradeoff between describing a concept problem clearly, and how strong the gender bias in the framing can be.

There are undoubtedly additional factors such as students' prior knowledge, exam anxiety or disability which intersect with gender and may have a greater impact on exam performance than assessment design. Our experiments could not control for all possible combinations of these factors, and further investigation of how their intersection impacts performance on assessment is warranted.

8 Conclusions and Future Work

This paper examined how gendered framings of real-world contexts in CS assessments relate to gender differences in exam performance. Across two controlled experiments with complementary designs, we found that gendered framings can affect student outcomes when they are applied consistently across an entire exam. Individual question framings may be too subtle to produce measurable effects in isolation, but their cumulative presence at the exam level can meaningfully shape performance. In our case we observed that a trial exam with a consistently masculine bias benefits men's grade performance. These results highlight the importance of assessment design as a structural component of equity in CS education, and support the view that gender gaps emerge through repeated interaction with biased learning environments.

As a next step, we plan to adopt the fully crossed exam design used in Study Two in an upcoming final exam. This design assigns the same tasks to all students, allowing us to investigate a balanced range of topic framings within the ethical constraints surrounding real exams. Additionally, we will revisit the prestudy to further examine how students perceive gendered topics, and to compare their responses with those of CS instructors. Finally, we will investigate the role of underlying social and psychological mechanisms that may mediate the observed effects, such as stereotype threat [24], sense of belonging [7], and test anxiety [12]. Understanding how these factors interact with assessment design could inform more systematic and inclusive approaches to assessment in CS education.

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